L2.1 Determinants (part 3)

Transcript Language

English

Hello, and welcome to the maths 2 component of the online BSc course on data science.

Today s video is a continuation of the determinants topic that we have been studying. So,

this is part three of that same topic. So, let us recall quickly what we did in the

previous so far in this topic. So, we have defined for a 1 by 1 matrix, that the determinant of the matrix a is just the number a. And then we have seen that for an n by n matrix, the

determinant is defined inductively. So, we did the definition explicitly for 2 by 2 and 3 by 3.

And then I showed by for 4 by 4 how I can use a 3 by 3 definition

and then you we said that, in general, we use the minors or co-factors, corresponding to the

first row in order to define the determinant. So, that means if so the idea is if you know

how to determine to define the determinant for n minus 1 by n minus 1 matrix, then you can use this

procedure to get it for the n by n matrix. So, just to recall, the i, j-th minor is the

determinant of the submatrix formed by deleting the i-th row and j-th columns, so you have

n rows and n columns, you delete the i-th row and the j-th column. And then you are

left with an n minus 1 by n minus 1 matrix. And then you take the determinant of that.

And the i, j-th cofactor is minus 1 to the power i plus j times the i, j-th minor.

And just to recall again, this was the final thing that we did in the previous video, we have seen that the determinant of a is minus 1 to the power 1 plus j, a1j, M1j.

So, this was given by what I called expansion along the first row and then you can take this

minus 1 to the power 1 plus j, inside along with M1j and then you can replace it by C1j

that is why we define the cofactors where we can get rid of the science coming here.

So, this is what we have studied so far. So, why did we study the i, j-th minor or the i,

j-th cofactor, because it is not occurred so far. So, that is one of the things we

will see in this video and we will also see some other properties of the determinant.

So, why did we define the i, j-th minor or the i, j-th cofactor? So, the reason is because

we had a formula for the determinant in fact the definition

in terms of expanding for any row or any sorry with respect to the first row.

And now what we are going to see is that you can actually do this with any row or any column.

And there is nothing very special about the first row. So, this is called expansion along any row or column. So, here is what happens when we do expansion along the i-th row.

So, in that case we can write the expression for the determinant as the sum of the

i, j-th entry of a times the i. j-th minor multiplied by minus 1 to the power i plus j.

So, remember, when i was 1 that was the definition. So, when i was 1, we had minus

1 to the power 1 plus j a1j, M1j. So, now what we are saying is we can replace that 1 by i and

the formula works equally well. So, the expansion along the first row was the definition that we are

going to use and what we are saying is, it does not matter whether you use one there or any other

row, you can expand along any other row. And the fact is, you can even use a column

to do the same thing. So, you can expand along any column. So, here, notice that

the j was what was varying and the i was fixed. So, this is expansion along the i-th row

and this is the expansion. So, now the j is fixed and the i is varying.

Here i is varying and j is fixed. So, that is expansion along the j-th column.

So, maybe let us do an explicit expression, let us write down one explicit expression here.

So, suppose I have a 3 by 3 matrix, and I want to expand along the second row.

So, what is this term telling me? So, this is expansion along second row.

So, this is saying what you have to do is, you have to take minus 1 to the power. So, now i is 2

and now j varies, so minus 1 to the power 2 plus 1 times a21 times the 2 1-th cofactor sorry the 2

1-th minor plus minus 1 to the power 2 plus 2, a22 times the 22-th minor plus minus 1 to the power

2 plus 3 a23 times the 23-th minor, my bad this should be 22 so that is the expression.

So, if you write it out, concretely, this is minus a21 times determinant of

so you have to drop the second row and first column. So, you get a12, a13, a32, a33

plus a22 times determinant of a11, a13, a31, a33 minus a23 times determinant of a11, a12, a31, a32. So, I hope it is clear what we mean by expanding along the i-th row this is

an example of how to do this, you can do this for an n by n matrix also. It is correct.

The proof is slightly involved I mean, it is not hard, but it involves manipulation of

a lot of expressions. So, if you like your algebra, I will suggest you go ahead. And

let me do maybe a similar example for expansion along the second row sorry second column. So,

this is expansion along second column. So, what this is saying is that this is minus 1 to

the power 1 plus 2, a12 times determinant of a21, a23, a31, a33 plus minus 1 to the power 2 plus 2

a22 times

similar determinant, I will encourage you to write this down plus minus 1 to the power 2 plus 3

sorry this was what the second column, so my bad, this is a32 times the determinant of

the corresponding matrix you obtain namely so that is a minor.

So, this is what we mean by expansion along any row or column. And it is a fact that the determinant can be computed in any of these ways. And it remains the same, so there is no

nothing special about the first row of the matrix. That is how we define it. So, now, that we have

seen that the determinant can be computed in terms of any row or column meaning by expanding along

any row or column, we see that in particular, there is nothing special about the first row.

And we see that the expression that we are getting, there is some kind of

symmetry about them. But of course, with some negative signs thrown in and so on.

So, keeping that in mind, let us, now look at some properties of the determinant. So, we already saw

in the first video, that the determinant of a product is the product of determinants and

we could do various operations on matrices and the determinant either remain the same or change

sign. So, we need all of those 2 by 2 and 3 by 3 matrices. So, let us see now what happens for

the general case. So, I will make these statements but I will not really prove too many of these.

So, the first property that we want to look at is the product. So, the same thing that

was earlier that we saw earlier still holds, namely that a determinant of a product is

the product of determinants. So, what we mean by that is determinant of AB

is determinant of A times determinant of B. So, we explicitly check this, when they were of

size 2 by 2 and I said you can try to check this for 3 by 3. If you did, you must have seen that

the expression start becoming rather complicated, it is not difficult, but you have to be careful

with your algebra. And so you can imagine that for 3 by 3, sorry, beyond 3 by 3, if you

work it out in terms of the explicit expressions, it going to be slightly hard.

So, I will not, we will not really get into the proof of the statement. But

we have seen it for 2 by 2, and hopefully we have taken for 3 by 3. So, this is believable.

So, let us use this instead and derive some interesting formulae. So, the first thing that we can do is we can ask, if you take the power of a matrix of a square matrix of course,

namely you do A square A cube A to the power 4, A to power n in general,

then what happens to the determinant. And what this identity tells you is that

you can take this power out, so determinant of A to the power n is determinant of

A to the power n. The second thing that you get out of this identity is if you take the

determinant of the inverse, then that is just the reciprocal of the determinant of A. So, I can

take this more fancy language as determinant of A to the power minus 1.

So, the proof of this is exactly the same as we did for the 2 by 2 and 3 by 3 case, I will

suggest that you go back and look at it. Another very useful

formula is that when you get determinant of P inverse AP, so you

have a matrix P, you take its inverse. So, A and P are both of size square matrices of size n by n,

and P has an inverse. So, take determinant of P inverse AP, then

you can use this formula and what we derive just above to get that the determinant is the same as the determinant of A. And then another thing to note

very easy fact. Remember that we saw that the product of two matrices may not be

even defined, if you do not, if you take the opposite order and even if it is, they may be of

completely different size and even if they are of the same size, AB and DA may not be the same

But the determinant must be the same because the determinant is the product of determinants.

This is of course remember if A and B are of square matrices of size n

by n and the final identity maybe one can write down. So, determinant of A transpose A.

So, maybe before I even do this, I would ask, what is determinant of A transpose? So,

determinant of A transpose. So, now given that, we can expand along any row or column,

so for A transpose instead of expanding along the first row,

we expand along the first column. So expand along the first column,

so I say expand along first column. So, I strongly suggest you do this and if you do that, the expression you are going to get will allow you to conclude that this is determinant of

A. So, I will leave this to you so along with so I will have I should use this through this plus

what I will call induction. Namely, assume that this formula is correct for

n minus 1 by n minus 1 matrices, that d transpose of a matrix has the same determinant, then expand

along the first column and in the minors, remember there are determinants of n minus 1 by n minus

1 matrices. So, think of those as transpose of something, and you will see that you can

equate these two expressions. So, once we know this determinant of A transpose is

determinant of A, so determinant of A transpose A is determinant of A squared, this might not

be very seem very interesting right now, but maybe at some later point, you might

find expressions of this type coming up. So, this long list of interesting

observations, let us go on to another property. So, again, we have check this property for

2 by 2 and 3 by 3 matrices or other I have checked it for 2 by 2 and I hope you have solve 3 by 3.

So, switching two rows or columns changes the sign of the determinant.

So, what do we mean by

switching two rows or columns, let me recall on that was, so if A is like this,

this is the i-th row, this is the j-th row. So, this is the ai1, ai2, ain, this is aj1, aj2,

ajn. So, we will switch these two rows to get this new matrix A tilde. So, now it is i-th row will be

aj1, aj2, ajn this is i-th row of A tilde and the j-th row will be ai1, ai2 up to ain this is the

j-th row. So, we have switched these two column rows, and everything else in A tilde is exactly

the same as it was for A, all other entries are exactly the same. So, then determinant of A tilde

is minus determinant of A, how do we get this? So, the again the idea is

expand along the i-th row and then use induction. So, expand along i-th row

and use induction. So, if you do that, you will see a minus sign coming up.

Let us, move on to the next property that we want to discuss, I should have said in the previous

slide that similarly, if you interchange columns you get the same result. So, adding multiples of a

row to another row leaves the determinant unchanged. So, what does this mean?

So, here is A, which was like in the previous slide and what is A tilde. So, let us say I add

t times the j-th row to the i-th row. So, then here I get the ai1 plus t times aj1, ai2 plus

t times aj2 and so on ain plus t times ajn and the j-th row remains the same aj1, aj2, ajn.

So, the only thing that has changed is the is the i-th row and what happens in that case, so the

claim is the determinant of A tilde is the same as the determinant of A.

So, again, one has to prove this. So, the proof is so one way of proving this is to

expand along the i-th row and then you will get two different expressions one as this

determinant away and the other is determinant of a matrix, which has two rows which are the same and then one will have to prove that is 0. So, that is how you get this.

So, one can do this also for columns. So, for columns, let us say if I want to add

the k-th column to the l-th, sorry t times l-th column to the k-th column, then we will get we

want k plus t times a1l, a2k plus t times a2l and so on ank plus t times anl this is the new k-th

column and the same thing will have work the determinant of A double tilde is determinant of A.

So, adding multiples of a columns to another column leaves the determinant unchanged.

So, then another property. So, again, we have seen this for 2 by 2 and I

left to you to check for 3 by 3, if you multiply a particular row by a constant t,

then the determinant gets multiplied by t. So, the same thing holds for columns, if you multiply

a column by t, then the determinant gets multiplied by t. So, what is this thing,

so you have your matrix A and let us say in A tilde you multiply the i-th row by t.

So, we have tai1 tai2 and so on tain everything else is the same. So, then what you do is you

expand using the i-th row so, if you expand using ith row you get t times ai1, maybe

for that there is a minus sign which I should have included, anyways sign I (())(22:21) forget.

So, determinant of A tilde, so expansion along the i-th row gives us summation minus 1 to the

power i plus j j runs from 1 through n. So, maybe let me do this proof so that

you get an idea of how to prove such things. So, t times aij Mij but now you can take t out and

so you get minus 1 to the power i plus j

aij, Mij. So, now so you have to be very careful here because when you write down the minus that

with respect to a particular matrix, so let me put this tilde here. So, this is the minor

the i, j-th minor of the matrix at A tilde. So, this is i, j-th minor of the matrix A tilde

but if you delete i-th row and whatever column then the remaining matrix remember it will may

have the same matrix. So, m tilde i, j for this particular row i is the same as Mij.

So, this is just t times summation j is 1 through n minus 1 to the power i plus j aij Mij where this

is i, j-th minor corresponding to the matrix A and now, this is exactly the expression for the

determinant of A. So, we have actually proved this. So, what is upshot? The upshot is that

if you multiply a particular row by t then in the determinant it comes out, the same thing holds for

columns. So, I will suggest that you check this so there you will have to expand along columns.

So, I will also put in a warning here because this is something

that often students make mistakes with so warning.

If you multiply the entire matrix by a constant, remember that we have

studied multiplying matrices by scalars. So, scalar multiplication matrices that means you

multiply each entry of that matrix by that scalar. So, if you do that,

and this is an n by n matrix, so, just to keep track of that this is an n by n matrix,

then this is like multiplying not just one particular row or one particular column by t, but every row or every column by t. So, you will pick up a t in the determinant

from each row. So, what you will get this t to the power n times determinant of A.

What is this n? This n is exactly

the size of A remaining the number of rows or number of columns in A. So, t to the power n

times determinant of A, you will not get t times determinant of A. So, be careful of this property,

it says you multiply a row by a constant t, then that t comes out in the determinant. If

you multiply the entire matrix by t, then t to the power n comes out of the determinant.

This is something students often make mistakes, so be careful with this.

So finally, let me end with some computational tips. So, if you have

a zero row or a zero column in a matrix, then the determinant is 0. How do you get this? We

expand along that particular row or column. The determinant of a matrix in which one row or

column is a linear combination of the other rows, respectively columns is 0.

Why is that? Because you can remember, we can add multiples of a row to another

row and the determinant remains unchanged. So, you keep adding the correct expressions of

the other rows to that particular row and you make it 0. So, once you make it 0, remember,

the determinant remained unchanged and then you have a zero row or column. And that is how you

get the determinant to 0. So, the second property we will be using later on. So, we will come back

to this if you did not understand this. The third one is what we just saw scalar

multiplication of a row by a constant t multiplies the determinant by t. And the

fourth is a general sort of tip, while computing the determinant, you can choose to compute it

using whatever row or column is most convenient. For example, as we saw in the first thing here,

we used that row or column which is 0. So, this is a general statement or it may be

of use so what are we done in this video, we have seen how to compute determinants. First of all, we define determinants for square matrices and then we saw that you can compute them

using expansion along any row or column in terms of minors and then we use that to

look at some other properties. So, most importantly, determinant of a product is

product of determinants they are very important. So, keep this in mind.

And then we also saw that if you do some particular operations,

namely adding two rows meaning adding a multiple of one row to another,

or multiplying a particular row by a constant, or swapping rows, we saw how the determinant behaves.

So, that is it for now. In the next video, we will see how to use determinants to compute

solutions to inverses sorry solutions to systems of linear equations.